MULT20015 Elements of Quantum Computing

Practice Class 9

Welcome to Practice Class 9 of MULT20015 Elements of Quantum Computing.

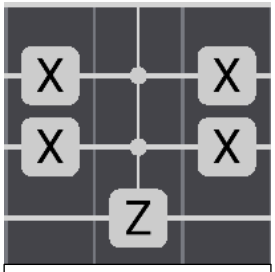
The purpose of this lab session is to:

- implement quantum counting algorithm
- implement Grover's algorithm for single solution case
- understand the concepts of QUBO

1 Implement quantum counting algorithm

In the lecture on Grover's algorithm we saw that the optimal number of Grover iterations (to maximise the probability of finding a solution) depends on the number of solutions relative to the size of the search space. In general, the number of solutions is unknown. Hence, we first need to compute the number of solutions.

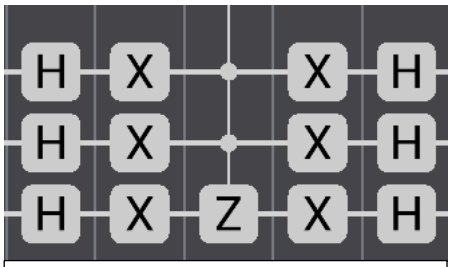
Please use the following circuit for a three qubit oracle in this exercise.



Oracle circuit

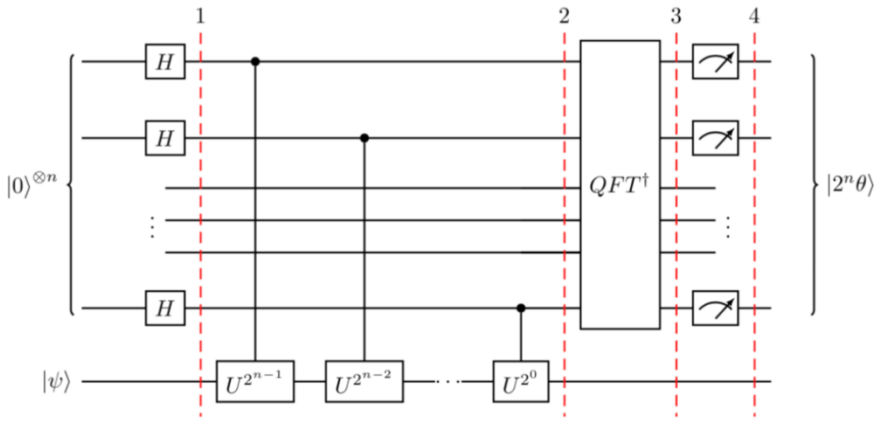
Note that though you know the details of the oracle, please take this as a given and do not count the number of solutions using the oracle.

Recall that a Grover iteration (we will use the term operation instead of iteration) requires two steps: (i) application of the Oracle, and (ii) inversion around the mean. A circuit for inversion around the mean is given below.



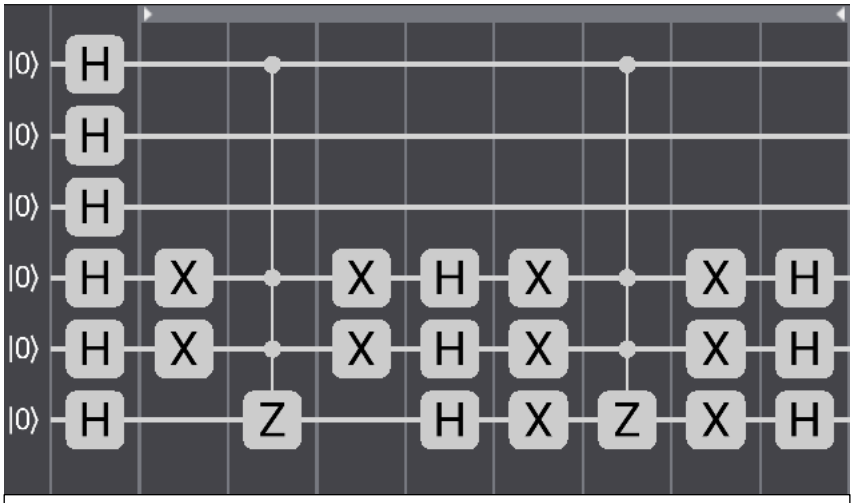
Inversion around mean

To count the number of solutions we combine Grover's algorithm with Phase estimation.



General phase estimation circuit

The operation U in the circuit will be Grover's operation (that is, using the oracle followed by inversion around the mean). Observe that to encode the controlled-Grover operation we only need to add an additional control on the Z gate. The circuit below depicts a controlled Grover operation.

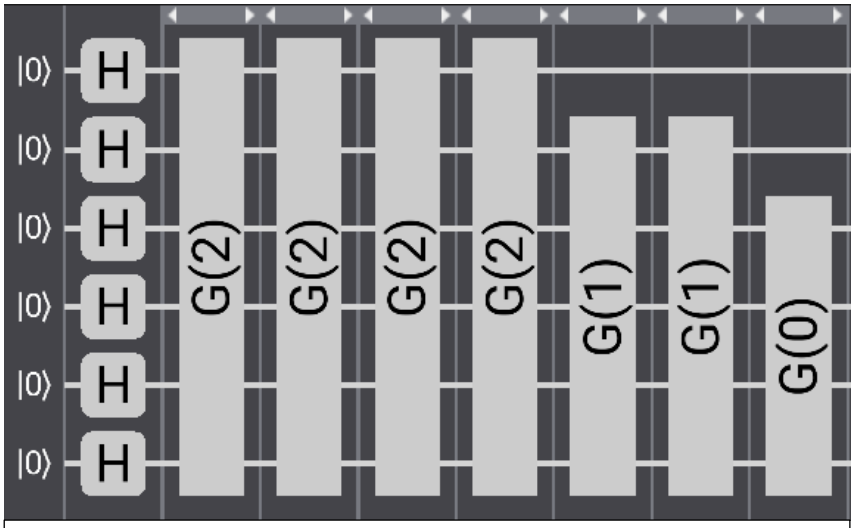


Controlled Grover operation (control on top qubit)

Exercise 1.1 Explain why one does not need controlled operation on the X gates in the circuit shown above.

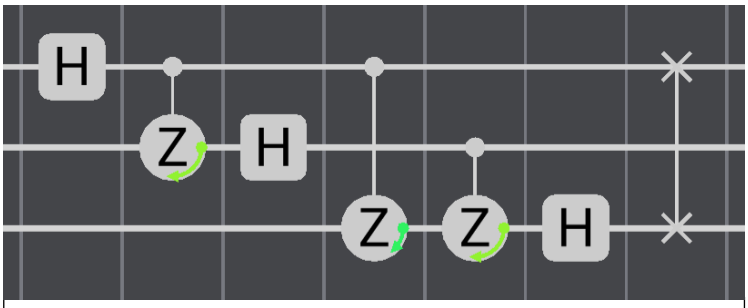
Exercise 1.2 Create a phase estimation circuit with 3 measuring qubits where the operation U is a Grover's iteration with the given oracle. The circuit will have two parts:

1. *Controlled Grover operations*. Since we are using 3 qubits for counting. We will need in total 4 Grover operations controlled over first qubit, 2 Grover operations controlled over the second qubit, and 1 Grover operations controlled over the third qubit. The circuit shown below depicts these operations.



Controlled Grover operations for 3 counting qubits

2. Inverse QFT. After your circuit with controlled Grover's operations, construct the Quantum Fourier Transform on the top three qubits. The circuit is shown below for your reference (it is the same as what you had used in the phase estimation lab).



Inverse QFT circuit

The angles for the controlled-Z rotations are as follows:

(a) $-\pi/2$ with a global phase of $-\pi/4$
(b) $-\pi/4$ with a global phase of $-\pi/8$
(c) $-\pi/2$ with a global phase of $-\pi/4$

Exercise 1.3 Take the state with the highest probability and extract the measured value for the phase estimation. Note that you will see probabilities for the overall state. To extract the measure value you have to exclude the qubits used for Grover's operation. For example, if the state is $|001010\rangle$, you should ignore the last three qubits to get the measured value of 1 (i.e., 001).

Exercise 1.4 Recall that if $U|\psi\rangle = e^{2\pi i\theta}|\psi\rangle$, the phase estimation circuit measures an estimate of $2^{\mathrm{n}}\theta$. The Eigenvalues for the Grover's operator are $e^{\pm 2i\theta}$. Calculate the angle θ for the Grover's operator.

Exercise 1.5 Using the obtained estimate of θ, calculate the number of solutions and optimal number of Grover's iterations for the given oracle.

2 Grover search

Now we'll put all that together to implement simple instances of Grover's search algorithm.

Exercise 2.1 Program a circuit in the QUI that implements Grover's search for the given oracle with the optimal number of iterations you calculated in the previous exercise.

Exercise 2.2 Revisit the oracle and verify that the number of solutions and the obtained solution.

3 Understanding QUBO

In this section we will explore the use of quadratic unconstrained binary optimisation problems. These will be used again next week when we explore how to use a quantum computer to solve these problems.

As you saw in lectures many optimisation problems over binary numbers can be cast into the "spin-glass" form where the task is to minimise the energy of a particular function E over variables $\{z_i = \pm 1\}$. The quadratic form (QUBO) involves products x_i *and* x_j, which we can map the problem onto a quantum computer by converting to variables z_i and z_j.

Exercise 3.1 If a QUBO problem has a term,

$$x_1 x_2$$

write down the corresponding energy term after we make the substitution,

$$x_i = \frac{1 + z_i}{2}$$

Exercise 3.2

We will work to solve a problem for which the quantum cost function (which we would like to minimize) has been determined to be:

$$H = Z_1 + 2Z_1Z_2 + Z_2Z_3$$

For each of the eight possible basis states (why eight?), fill in the following table, evaluating the cost function for each basis state:

State, i	Z_1	Z_2	Z_3	H_i
$\|000\rangle$	+1	+1	+1	4
$\|001\rangle$	+1	+1	-1	2
$\|010\rangle$				

$\|011\rangle$				
$\|100\rangle$				
$\|101\rangle$				
$\|110\rangle$				
$\|111\rangle$				

Exercise 3.3

Which state has the minimum cost?

Finally, let us consider an instance of the number partitioning. The problem is stated as follows: given a set S of numbers $\{w_i\}$, is there a partition of this set of numbers into two disjoint subsets R and S – R, such that the sum of the elements in both sets is the same? In lectures we saw the solution is equivalent to finding the basis state with the minimum energy in the system given by the "Hamiltonian" (quantum cost function):

$$H = \left(\sum_i w_i Z_i\right)^2 = \sum_{i \neq j} 2 w_i w_j Z_i Z_j + \sum_i w_i^2$$

where and each number is assigned a qubit ($w_i \leftrightarrow q_i$) and the qubit value 0 or 1 indicates which of the two sets, R and S–R, the numbers are associated with.

Exercise 3.4 Consider $S = \{1, 2, 3\}$. Classically solving the problem (i.e. in your head), what are the solution sets R and S–R?

Exercise 3.5 Convert classical solutions to quantum land. Assigning "R" ↔ "0" and "S–R" ↔ "1", translate the expected possible solutions from 2.1 into qubit form $|q_1 q_2 q_3\rangle$ where that values of each q_i = 0, 1 indicates which set the corresponding number w_i is assigned to in the solution. The two solutions are:

_____ and _____

Exercise 3.6 The Hamiltonian of this number partitioning problem is

$$H = 4Z_1 Z_2 + 6Z_1 Z_3 + 12 Z_2 Z_3 + 14.$$

Make sure you understand how to get this from the general expression above. Show that the energy of this system is zero for the solution basis states $|q_1 q_2 q_3\rangle$ you found in 3.5, and non-zero for other states.